LIFE ON BOARD

DISEASE, MEDICINE, AND SURGERY

THROUGHOUT MOST OF the history of naval warfare, disease took a far heavier toll on sailors' lives than actual combat. Until the 18th century there was little any navy could do about it, because medical knowledge remained too primitive. Gradually, common-sense application of principles of hygiene and nutrition by good captains became systematized to good effect. Treatment of those injured in battle also remained a crude business until modern times.

SICKLY SAILING SHIPS

Although conditions on board galleys were often unhygenic and overcrowding chronic, galleys stayed at sea for only short periods at one time. They suffered epidemics like any accumulation of men in pre-modern times—the Venetians lost 20,000 crew in their fleet, probably to typhoid fever, in June 1570. But once sailing ships started going to sea for long periods death rates regularly soared. It was common for ships on oceanic voyages to lose a third to a half of their crew to disease. A lengthy operation such as a blockade was virtually impossible because disease inevitably decimated the crews after a time. Large numbers of men were cooped up together in overcrowded conditions, in humid, unventilated spaces between decks, with food supplies short on freshness and variety. Scurvy was a typical sailor's disease. Typhus—often brought on board by recruits from jails—and tuberculosis were great killers, as was yellow fever in the tropics.



LEMONS During the Napoleonic Wars the Royal Navy began to systematically carry citrus fruit on board ship to prevent scurvy, a disease caused by lack of vitamin C.

> *When a patient dies from a local injury, with the limb on, we are apt to blame ourselves . . . but it does not follow that the operation would have saved him.*
>
> **Lancelot Haire,** *Remarks on Mr Lucas's practical observations on amputations*, 1786

MEDICINE A supply of medicines, ointments, and instruments was to hand in the sick berth on board a British ship like HMS *Victory*. A larger lockable dispensary was situated on the orlop level where the surgeon, as a professional man, had his own cabin.



SHIPS' DOCTORS

Medical staff on ships were rarely properly trained. A barber surgeon on a sailing ship would be expected to deal in a rough-and-ready way with a steady stream of fractures and hernias gathered through the heavy and hazardous nature of the work on board, as well as venereal disease after a spell in port. His real challenge, of course, came in battle. Men who might survive mostly had limbs shattered by solid shot or gouge wounds caused by flying splinters of wood. Amputations were performed in two minutes, the limbs collected in a basket by the surgeon's assistant, the loblolly man. Speed was the most appreciated quality as there were no anaesthetics except alcohol. Surgery was performed below the waterline, if the ship was big enough to permit it, so as to be safe from shot. Wounds almost always became infected, but various primitive techniques ensured reasonable survival rates.



OPERATING TABLE The surgeon's operating room in HMS *Victory* was in the cockpit, an open area near his cabin, located on the orlop deck well below the waterline.



SURGEON'S TOURNIQUET The 19th-century Petit type of screw tourniquet (the strap is missing) was used to stem bleeding as patients waited for the surgeon to operate.



SURGEON'S SAW Tenon saws, such as this one, were used for amputation on British ships, since the stout blades were less likely to break. Amputation was a last resort, and speed and efficiency were required from a good surgeon.